Provider-Dependent Reliability of Self-Reported Confidence under a Shared Black-Box MT Protocol

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Abstract

This paper studies a trustworthy-AI reliability question for black-box language-model services: can one shared confidence threshold be reused across providers under a common protocol? We evaluate English→German translation with eight models from OpenAI, Anthropic, and Google on 500 WMT17 sentences (4,000 translation–confidence pairs) at temperature 0. Quality is scored with sentence-level chrF, and low quality is operationally defined as the within-model bottom 20% tail. Relative to that target, expected calibration error (ECE) and mismatch@0.9 differ widely despite similar mean chrF (54.0–60.1): ECE ranges from 0.076 to 0.180, and mismatch@0.9 ranges from 0.2% to 18.2%. The released snapshot also shows uneven parse-warning burden (443/4000 rows; 11.1%), an implementation-level confound for cross-provider reliability comparisons. The central claim is therefore practical and scoped: under this shared protocol and operational target, self-reported confidence is provider-dependent and non-portable as a common probability scale, so a fixed threshold does not induce comparable routing/abstention operating points. Supplementary analyses indicate that provider-specific isotonic remapping can reduce ECE but does not restore portability. Limitations are explicit: operational (not semantic) target, no bundled COMET/xCOMET checkpoints, no completed semantic-audit labels, and no bundled non-baseline prompt-variant outputs.

1 Introduction

We study whether prompt-elicited self-reported confidence can be reused across providers as a common threshold interface for MT routing/abstention decisions. The analysis is deliberately operational: low quality is defined by reference-overlap tails, not human semantic labels. The question is practical rather than theoretical: does one threshold (e.g., 0.9) induce comparable operating points across providers under a shared protocol?

Across eight models, mean chrF is relatively clustered while confidence behavior is not. This disconnect creates portability risk: identical threshold policies can produce materially different high-confidence failure rates. We therefore treat confidence as a provider-specific decision signal unless calibrated per provider.

2 Related work

MT reliability work spans reference-based evaluation (e.g., BLEU/chrF), learned metrics, and QE without references [10, 11, 12, 5, 13]. This paper does not propose a new metric or QE model; it asks whether the generating model's own confidence is stable enough for cross-provider threshold transfer [1].

In LLM settings, confidence and calibration have become central for selective prediction and abstention [3]. Prior work shows miscalibration in modern models and motivates post-hoc correction methods [6, 7, 14]; black-box settings are particularly challenging because token-level internals are unavailable [15].

Model	Mean chrF	ECE (within-q20)	Mismatch@0.9	Median conf.
anthropic/claude-haiku-4-5-20251001	57.01	0.094	13.4%	0.92
anthropic/claude-opus-4-6	58.45	0.087	7.6%	0.93
anthropic/claude-sonnet-4-5-20250929	56.89	0.095	12.8%	0.95
gemini/gemini-2.5-flash	54.01	0.148	4.2%	0.73
gemini/gemini-2.5-pro	60.07	0.180	18.2%	1.00
openai/gpt-5-mini	57.09	0.152	12.8%	0.93
openai/gpt-5-nano	55.31	0.076	0.2%	0.73
openai/gpt-5.2	58.87	0.134	15.2%	0.93

Table 1: Main summary table from regenerated code-side outputs (within-model chrF-q20 target).

Recent work on verbalized/self-reported confidence indicates useful signal but also sensitivity to elicitation and imperfect uncertainty expression [16]. Our contribution is a constrained MT portability test: same task, same protocol, multiple providers, and operational chrF/BLEU targets.

3 Methods

Setup. We evaluate WMT17 en→de with a fixed 500-sentence sample (seed 123) [2]. Eight models (OpenAI/Anthropic/Google) each produce one translation and one scalar confidence in $[0, 1]$ per sentence (4,000 outputs total). Canonical prompts and JSON schema are shared; provider adapters only handle transport.

Operational target. Sentence-level chrF is the primary quality signal [11]. Low quality is the within-model bottom 20% chrF tail. This is an operational overlap-based target, not semantic correctness calibration. A secondary robustness stage uses the strongest available alternative metric in the environment (BLEU fallback when COMET is unavailable) [10, 12].

Metrics. With 10 bins, ECE is computed against the binary non-low-quality label, following standard calibration usage of ECE in modern ML [6]:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|S_b|}{N} \left|\mathrm{acc}(S_b) - \mathrm{conf}(S_b)\right| .$$

Mismatch at threshold τ is

$$\Pr(\text{low-quality} \wedge \text{confidence} > \tau),$$

reported mainly at $\tau = 0.9$ (mismatch@0.9), a strict policy-relevant regime.

Complexity proxy. We include a lightweight source-side surface-complexity proxy (length/depth-like/rarity/name features) only for secondary stratification; it is not a validated semantic-difficulty measure.

Implementation details, full prompts, and script-level provenance are in the repository.

4 Results

Table 1 shows the core result. Mean chrF is similar (54.0–60.1), but ECE (0.076–0.180) and mismatch@0.9 (0.2%–18.2%) vary widely. Thus a fixed threshold does not transfer cleanly across providers.

Provider-specific isotonic remapping can reduce ECE for several models on held-out data, but gains are uneven and not portable; one model worsens in held-out ECE. This follows classic

Model	ECE before	ECE after	Mismatch@0.9 before	Mismatch@0.9 after
anthropic/claude-haiku-4-5-20251001	0.110	0.027	14.3%	0.3%
anthropic/claude-opus-4-6	0.084	0.114	9.0%	2.1%
anthropic/claude-sonnet-4-5-20250929	0.087	0.073	12.1%	1.0%
gemini/gemini-2.5-flash	0.152	0.055	4.3%	0.3%
gemini/gemini-2.5-pro	0.175	0.087	17.5%	0.0%
openai/gpt-5-mini	0.128	0.116	10.8%	0.0%
openai/gpt-5-nano	0.091	0.045	0.3%	0.3%
openai/gpt-5.2	0.155	0.047	16.3%	0.0%

Table 2: Held-out isotonic calibration summary from regenerated calibration artifacts.

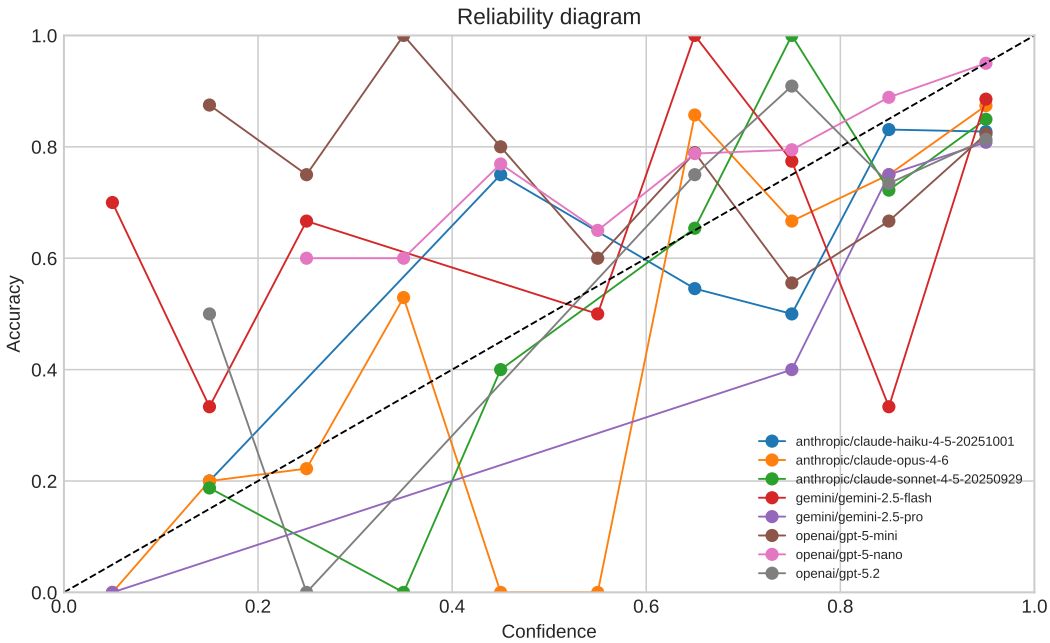

Figure 1: Reliability diagram (qualitative) under the operational chrF-based label.

post-hoc probability calibration practice where isotonic mapping is model-specific [9]. This supports mitigation as provider-specific, not universal portability restoration.

Prompt-variant infrastructure and semantic-audit scaffolding are available in the repository, but non-baseline prompt outputs and completed semantic labels are not bundled in this release.

5 Discussion and limitations

The main finding is comparative and operational: under one shared protocol, self-reported confidence behaves as a provider-dependent score, so fixed thresholds induce different routing/abstention points. This does not imply semantic calibration.

Limitations remain explicit: (i) target is operational chrF/BLEU overlap, not semantic correctness; (ii) no completed human semantic-audit labels; (iii) no bundled non-baseline prompt-variant outputs; (iv) one language pair/dataset with 500 sentences; (v) parse-warning burden (443/4000) is an implementation-level confound.

Full prompts, mismatch examples, robustness exports, and artifact-level provenance are included in the accompanying repository.

6 Conclusion

Across 500 WMT17 sentences and 4,000 outputs from eight LLMs, confidence portability fails under a shared black-box MT protocol: similar mean chrF coexists with large cross-provider gaps in ECE and mismatch@0.9. The central claim is practical and scoped to this operational target.

Provider-specific isotonic calibration may mitigate alignment within providers, but it does not restore a shared probability scale across providers. Deployment should therefore use provider-specific threshold tuning/calibration or stronger external evaluator layers rather than assuming universal threshold portability; stronger learned evaluators are a natural next layer [8, 4].

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